

Ashton Honnecke

Senior Software/Cloud Engineer • Python • AWS • DevOps

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HIGHLIGHTS

- Self-motivated engineer focused on maintainable, reusable code and pragmatic delivery.
- Operates comfortably from low-level systems detail to high-level business outcomes.
- Clear communicator (written & verbal); results-driven and collaborative.
- Experience running/serving extremely high-scale, public-facing web properties.
- Speaker: PyColorado — Cleaning Up Your Python Environment: Superfund Site.
- Speaker: PyDEN — The Value of Linting.

SKILLS

Software development across multiple languages, frameworks, and stacks (~25 yrs).

Linux/Unix administration (~20 yrs); DevOps (cloud & colo) (~10 yrs).

Startups (founder & early-stage) (~15 yrs), including profitable exits or profitability (~10 yrs).

AI/LLM-native engineering (production LLM extraction pipelines, agent-directed development, multi-provider orchestration) (~1 yr production, ongoing).

Blockchain (ETH/BTC nodes, data engineering) (~2 yrs).

EXPERIENCE

CrewCapable AI — Chief Technology Officer

2025–Present

Subject Matter Expert: TypeScript, Python, Next.js 15, React 19, AWS CDK, Lambda, Step Functions, Postgres/Supabase, LLM extraction, Zod, Vitest, LocalStack, EDI X12

- Greenfield AI-native SaaS for corrugated packaging purchase-order intake: OCR + LLM extraction + human-in-the-loop review + automated ERP/EDI export; production pilot in under 10 months.
- Fully serverless AWS architecture on CDK: 10 Lambda functions; Step Functions workflow orchestrating OCR/extraction/reconciliation; event-driven integration across S3, Textract, EventBridge, SNS, Secrets Manager.
- Split infrastructure into 4 independently deployable CDK stacks (core backend, audit, ERP, management) to isolate blast radius and enable per-tenant rollout.
- Next.js 15 / React 19 frontend backed by ~86 typed API routes; Postgres (Supabase) with row-level security on every tenant-scoped table.
- Multi-provider LLM document-extraction pipeline (OpenAI, Gemini, Mistral, Tesseract, Texttract) with Zod-typed schemas; immutable audit trail separating raw model output from human-reviewed values.
- Document-processing state machine handling PDF, DOCX, image, and email-attachment intake with explicit failure and timeout paths.
- ~112k LOC TypeScript behind 94 test files (Vitest, Testing Library, LocalStack integration, ERP golden-file contract tests); 8 GitHub Actions workflows covering type checks, migration linting, ERP regression.
- Strict end-to-end type safety: branded domain types, Zod validation at every boundary, generated Supabase types; Biome + ESLint + Ultracite enforced in CI.
- 123 forward-only Postgres migrations with automated migration linting in CI to prevent environment drift.
- Per-Lambda CloudWatch alarms, state-machine failure alerts, commit-SHA logging on Lambda cold start, /api/version endpoint for incident-to-revision traceability.
- AWS Secrets Manager + per-Lambda IAM scoping + RLS on every customer-data table — no shared service accounts across tenants.
- Production integrations: Amtech DAT (SFTP), Epicor P21, QuickBooks, outbound X12 850 EDI; Stripe billing, Supabase Auth with Google OAuth, PostHog analytics, SES-based email intake.
- Turborepo / pnpm monorepo spanning web, backend, and 5 shared packages with LLM-optimized single-concern file markers.

Panasonic North America — Lead Cloud Engineer

2020–2025

Subject Matter Expert: Python, boto3, Bash, AWS (Lambda, SQS, SNS, Fargate, S3, CloudWatch, Cost Explorer), SonarQube, Docker, V2X/OBU telematics, SAE J2735 (DSRC), edge devices

- Chaired cross-team best-practices workgroup (~50 engineers) on OO, SOLID, code quality, and security; mentored developers across the org.
- Authored ADRs for event-driven/serverless and long-running serverful (Fargate) cloud-native architectures.
- Authored end-to-end incident playbooks: classification → response → resolution → blameless post-mortem.
- Built a near-real-time OBU ingress pipeline on Lambda/SQS/SNS with horizontal scale; worked with SAE J2735 DSRC messages in the data path; provisioned for throughput headroom not exercised in production.
- Shipped V2X messaging deployment for a Utah snowplow fleet; worked on edge devices, packet sniffing, and hardware↔cloud integration.
- Designed and released internal developer tooling as open-source on PyPI: hot Lambda filesystem mounting (Consolo), CLI SNS inspection (Snifter), Jira↔Git forge integration (Jolly Brancher).
- Delivered artifact build pipeline reducing build time and cost.
- Identified and decommissioned unused AWS resources via Cost Explorer and tagging discipline → up to ~30% infrastructure savings.
- Drove self-hosted SonarQube rollout across the org → >50% unit-test coverage gains across multiple languages.

Digital Assets Data — Sr. Software Engineer / Blockchain Tech Lead 2018–2020

Subject Matter Expert: Python, Django, React, Pandas, Docker, AWS (Lambda, API Gateway, S3, CloudWatch, ECS), Google BigQuery, CircleCI, BTC/ETH nodes, smart contracts

- Built Python microservices to extract on-chain BTC/ETH data directly from cloud-hosted nodes (low-latency ingest; pandas; scheduled CloudWatch Events).
- Implemented Lambda + API Gateway microservices for cryptographic decoding at very high throughput.
- AWS ingestion pipelines on S3, CloudWatch Events, Lambda, and ECS — processed 100 GB nightly with individual files up to 30 GB.
- Worked with Google BigQuery to extract, normalize, and ingest Bitcoin and Ethereum datasets.
- Implemented QR-code TOTP multi-factor authentication in Django/React; hardened and containerized blockchain nodes across per-env configurations.
- "Fan-out" CircleCI workflows to parallelize PR tests; standardized unit testing + linting; containerized every stack with Docker.
- Automated external-library dependency PRs with integration checks against external APIs in the PR test flow only.
- Tech lead with 1 direct report (blockchain team of 2 at peak); owned on-call playbooks (VictorOps, OpsGenie); performed many interviews.
- Built centralized unhandled-exception capture/logging and uptime monitoring across multiple projects and languages.
- Restructured deploys into a single "push-button" script; wrote and distributed git hooks for linting, typing, and stylistic cohesion.

Havenly — Senior Software Engineer 2015–2017

Subject Matter Expert: PHP, CakePHP 2/3, Linux, Git, SQL, queued-ingest systems, data modeling

- Most senior engineer on the data-ingest initiative within an ~8-person team; designed and shipped a greenfield, horizontally scalable queued data-feed system: 7M+ rows/day from 25+ sources in 15 formats.
- Built decoupled pipeline: raw retention (traceability/replay) → transform/validate/store → push to external API with faceted fuzzy search.
- Authored CakePHP2 → CakePHP3 migration plan; refactored into OO Components and Behaviors (CakePHP's mixins/traits) to enable encapsulation and unit testability.

Colorado State University Global Campus — Systems Architect (pro tem)

2014–2015

Subject Matter Expert: Linux, PHP, SQL, data modeling, SDLC, pentesting, Git, SSH

- Covered a vacated architecture-leadership role at CSU Global (director-level scope); led 1 engineer, owned architecture and SDLC for student portal, course registration, and transcript systems.

Pixelstub Consulting — Founder 2009–Present

Subject Matter Expert: Linux, PHP, JavaScript, SQL, SDLC, pentesting, Git, SSH, LLM extraction pipelines

- Founder of a long-running independent consultancy (profitable when actively engaged); broad tech-stack engagements across web, mobile, data, and DevOps.
- Built an LLM-driven image-to-structured-data pipeline for menu data ingestion (picture → structured records).

Photobucket — Software Engineer → Senior Software Engineer 2005–2009

Subject Matter Expert: PHP, MVC frameworks, PayPal API, credit-card payments, trust & safety tooling, AJAX, CI, MySQL

- W-2 engineer at Photobucket during its peak-traffic era (widely reported as top-50 global web property, ~100M users); promoted SWE → Sr SWE based on full ownership of internal tooling.
- Built trust & safety reporting pipeline: tool that packaged offensive content and transmitted to the National Center for Missing & Exploited Children (NCMEC); contributed to multiple predator arrests via proactive law-enforcement integration.
- Owned payments (PayPal + credit-card processing, reconciliation) and account-management tooling used for legal/compliance archival.
- Created a Propel-like model code-generation system with unit tests; migrated raw PHP pages to an MVC framework for internal tools.
- Introduced CI and unit testing into the team's workflow; mentored engineers on design, flow, and integration.
- Built an AJAX-based status/exception display system for consistent UX, surfacing caught DB exceptions.

SPEAKING

Cleaning Up Your Python Environment: Superfund Site
PyColorado

[Slides and Code](#) [Video](#)

The Value of Linting
PyDEN

[Materials](#)

EDUCATION

Master of Business Administration
University of Denver · 2004–2005
Emphasis: Accounting

Bachelor of Computer Science
University of Denver · 2000–2004
Minors: Mathematics, Philosophy

OPEN SOURCE

Console

Hot-mount Lambda filesystems for rapid iteration; "hot reloading."

Snifter

Fast SNS topic/message inspection from the CLI.

Jolly Brancher

Git/Jira helper: find tickets, create matching branches, enforce branch hygiene.

INTERESTS

I enjoy snowboarding, cooking, board games, reading, welding, electric bass, and a steady stream of hands-on projects. I love puzzles and problem-solving and contribute to open-source projects.